

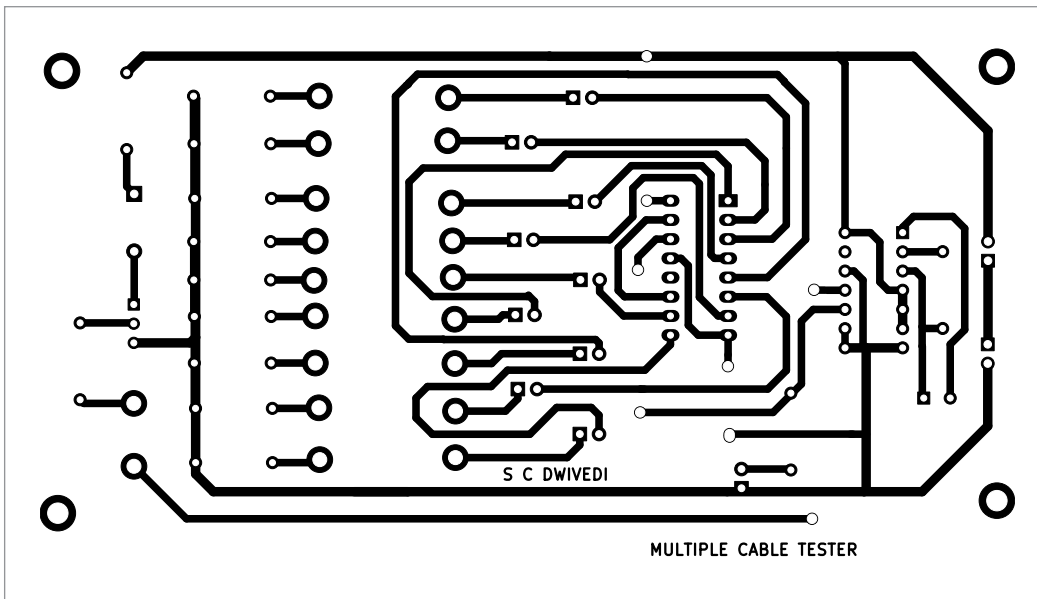


Fig. 3: Actual-size PCB of multiwire tester

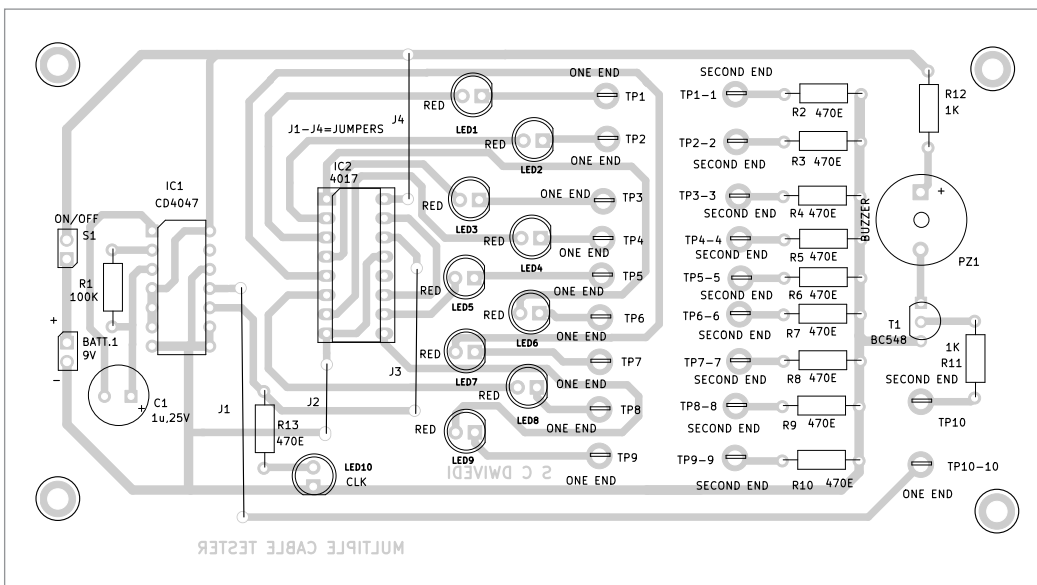


Fig. 4: Components side of PCB

This allows current to flow through the probes and into the base of the transistor, causing T1 to conduct. As a result, a larger current flows from the collector to the emitter, passing through the piezo-buzzer (PZ1) and producing an audible sound. The buzzer confirms continuity, indicating that the wire or the PCB track has no breaks.

If no current flows through the probes due to a disconnection or break, the transistor remains cut off, and the buzzer remains silent, indicating a broken circuit.

Construction and testing

An actual-size, single-sided PCB layout for the multiwire continuity tester is shown in Fig. 3, with

its component layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box. Mount LED1 through LED10 and switch S1 on the front panel.

Fix a 9-pin connector to the front panel and connect it to the PCB, labelling it TP1 through TP9. Similarly, fix another 9-pin connector 20-30 centimetres away, also connected to the PCB, and label it TP1-1 through TP10-10.

To check continuity, connect the nine-wire cable between TP1 through TP9 and TP1-1 through TP9-9. Mount switch S1 on the front panel at a convenient position for turning the circuit on and off. When switch S1 is turned on and the 9V battery is connected, LED10 should blink, indicating that the

circuit is powered. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/live-diy-multiwire-continuity-tester>

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